Units 4, 5, 6
Data Manipulation

1.

			Description
00	LOAD R0, 00	20 00	Load the value '00' into register R0.
02	LOAD R1, 2A	11 2A	R1 = memory[2A].
04	LOAD R2, 01	22 01	Load the value '01' into register R2.
06	XOR R2, R1, R2	92 12	XOR the value in R1 with R2.
08	JUMP R2, 0E	B2 0E	If R2 equals '00', jump to address 0E.
0A	JUMP R0, 2C	B0 2C	Unconditionally jump to address 2C.
0E	LOAD R2, FF	22 FF	Load the value 'FF' into register R2.
10	STORE R2, 27	32 27	Store 'FF' into memory location 27.
12	HALT	C0 00	End of the program.
2C	LOAD R1, 00	21 00	Load the value '00' into register R1.
2E	STORE R1, 27	31 27	Store '00' into memory location 27.
30	LOAD R1, 7F	11 7F	R1 = memory[7F].
32	LOAD R2, 00	22 00	Load the value '00' into register R2 (result accumulator).
34	LOAD R4, 80	24 80	Load the value '80' into register R4 (position mask).
36	LOAD R5, 08	25 08	Load the value '08' into register R5 (loop counter).
38	JUMP R5, 4C	B5 4C	If R5 equals R0, jump to address 4C.
3A	LOAD R3, 01	23 01	Load the value '01' into register R3 (bit mask).
3C	AND R3, R1, R3	83 13	AND R1 with R3, extract the LSB of source into R3.
3E	JUMP R3, 42	B3 42	If R3 equals R0, jump to address 42.
40	OR R2, R2, R4	72 24	Set bit in R2.
42	ROTATE R4, 01	A4 01	Rotate R4 right by 1 bit (shift position mask).
44	ROTATE R1, 01	A1 01	Rotate R1 right by 1 bit (get next source bit).
46	LOAD R3, FF	23 FF	Load the value 'FF' into register R3.
48	ADD R5, R5, R3	55 53	Add R5 and R3, decrement the counter by 1.
4A	JUMP R0, 38	B0 38	Unconditionally jump back to address 38.
4C	STORE R2, 7F	32 7F	memory[7F] = R2.
4E	HALT	C0 00	End of the program.

2.

- R0 = 0, R1 = 0, R2 = 0, R3 = 0, …

	PC	IR	Instruction	Round 1	Round 2
1	1E	14 24	LOAD R4, 24	R4 = MEM[24] = 11	
2	20	10 26	LOAD R0, 26	R0 = MEM[26] = 12	
3	22	23 2F	LOAD R3, 2F	R3 = 2F	
4	24	11 2F	LOAD R1, 2F	R1 = MEM[2F] = 40	R1 = MEM[2F] = 40
5	26	12 25	LOAD R2, 25	R2 = MEM[25] = 2F	R2 = MEM[25] = 30
6	28	A0 20	ADD R2, 01	R2 = 30	R2 = 31
7	2A	32 25	STORE R2, 25	MEM[25] = 30	MEM[25] = 31
8	2C	53 13	ADD R3, R1, R3	R3 = 6F	R3 = AF
9	2E	A0 40	ADD R4, 01	R4 = 12	R4 = 13
10	30	B4 24	JUMP R4, 24	R4 == R0	R4 ≠ R0
11	32	C0 00	HALT		Program Halts

R0	R1	R2	R3	R4
$(12)_{16} = 18$	$(40)_{16} = 64$	$(31)_{16} = 49$	$(AF)_{16} = 175$	$(13)_{16} = 19$

Ans: 175